Report on Project

ML-MEDICAL INSURANCE COST PREDICTION

Course Code: CS2003 - 1

Course Name: Machine Learning

Semester: VI SEM   Section: C

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Date of submission: 27-03-2026

Signature of Course Instructor

CERTIFICATE

This is to certify that Mr. Rishab Raj (USN: NNM23CS154) and Mr. Samarth Venkappa Patil (USN: NNM24CS515), bonafide students of III Year B.Tech in Computer Science and Engineering, NMAM Institute of Technology, Nitte, have successfully completed a Mini Project titled "ML-Based Medical Insurance Cost Prediction" during the academic year 2025–2026.This project has been carried out in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

Name and Signature of Course Instructor

Signature of HOD

ABSTRACT

This project focuses on predicting medical insurance costs using machine learning techniques based on various personal and health-related attributes such as age, gender, Body Mass Index (BMI), number of children, smoking habits, and residential region. The primary goal is to build an intelligent system that can accurately estimate insurance charges by analyzing historical data patterns.

A dataset containing medical insurance records is used to train the model. Before training, data preprocessing techniques such as handling categorical variables, encoding, and feature selection are applied to improve model performance. A **Random Forest Regressor** is utilized due to its ability to handle complex, non-linear relationships and provide high prediction accuracy.

The system is implemented as an interactive web application using Streamlit, allowing users to easily input their personal details and receive instant predictions of their expected insurance costs. The application also evaluates the model using performance metrics such as R^2 score and Root Mean Square Error (RMSE) to ensure reliability.

In addition to prediction, the system provides data visualization features to help users understand the distribution and trends of insurance charges. A unique feature of this project is the automatic generation of a detailed PDF report that includes user inputs, predicted cost, model performance metrics, and graphical analysis.

This project demonstrates the practical application of machine learning in the healthcare and insurance domain, helping users make informed financial decisions and enabling insurance providers to assess risk more effectively. It also highlights how data-driven solutions can improve efficiency, accuracy, and user experience in real-world applications.

TABLE OF CONTENTS

INTRODUCTION

In today's world, healthcare expenses are increasing rapidly due to lifestyle changes, environmental factors, and rising medical costs. Medical insurance plays a crucial role in providing financial protection against unexpected healthcare expenditures. However, determining the appropriate insurance cost for an individual is a complex process that depends on multiple factors such as age, health condition, lifestyle habits, and geographic location.

Traditionally, insurance companies calculate premiums based on manual analysis and predefined rules, which may not always provide accurate or personalized results. With the advancement of technology and the availability of large datasets, machine learning has emerged as a powerful tool to analyze complex data patterns and make accurate predictions.

This project, **"Medical Insurance Cost Prediction Using Machine Learning,"** aims to develop an intelligent system that can estimate insurance charges based on user-specific details. The system uses a dataset containing historical records of individuals and their corresponding medical expenses. By analyzing this data, the machine learning model learns the relationship between various input features and the insurance cost.

Among these, smoking habits and BMI are particularly significant as they directly impact an individual's health risk and medical expenses.

The project utilizes a **Random Forest Regressor**, a powerful ensemble machine learning algorithm known for its accuracy and ability to handle non-linear relationships. The model is trained and tested on the dataset to ensure reliable predictions. Performance metrics such as R^2 score and Root Mean Square Error (RMSE) are used to evaluate the effectiveness of the model.

To make the system user-friendly and accessible, a web-based interface is developed using **Streamlit**. This interface allows users to input their details easily and obtain instant predictions of their medical insurance cost. The application also includes data visualization features, enabling users to understand trends and distributions within the dataset.

An additional feature of this project is the generation of a **PDF report**, which provides a comprehensive summary of the user inputs, predicted insurance cost, model performance, and graphical analysis. This enhances the usability of the system for documentation and reporting purposes.

Overall, this project demonstrates how machine learning can be effectively applied in the healthcare and insurance domain to improve prediction accuracy, reduce manual effort, and support better decision-making. It highlights the importance of data-driven approaches in solving real-world problems and provides a foundation for further advancements in predictive analytics.

PROBLEM STATEMENT

In the modern healthcare industry, determining accurate medical insurance costs is a critical and challenging task. Insurance companies must evaluate multiple factors such as age, lifestyle, medical history, and environmental conditions to estimate the premium for an individual. However, traditional methods of calculating insurance costs are often based on manual rules, static formulas, and limited analysis, which may lead to inaccurate predictions and unfair pricing.

One of the major problems is the lack of personalization in insurance cost estimation. Individuals with different health conditions and lifestyles may be charged similar premiums due to generalized calculation methods. For example, a person with a healthy lifestyle may end up paying nearly the same insurance cost as someone with high-risk habits such as smoking, which is not an efficient or fair approach.

Another challenge is the increasing complexity and volume of healthcare data. With a large number of influencing factors such as Body Mass Index (BMI), number of dependents, smoking habits, and geographical location, it becomes difficult for traditional systems to analyze and identify meaningful patterns. This often results in inaccurate cost predictions and inefficient decision-making.

Furthermore, there is a lack of user-friendly tools that allow individuals to estimate their insurance costs before purchasing a policy. Most existing systems do not provide transparency or instant feedback, making it difficult for users to understand how different factors affect their insurance charges.

In addition, insurance providers face challenges in risk assessment. Incorrect estimation of insurance costs can lead to financial losses for companies or overpricing for customers, affecting both business performance and customer satisfaction.

OBJECTIVES

The primary objective of this project is to design and develop an intelligent system that can accurately predict medical insurance charges using machine learning techniques. The project aims to solve real-world problems in the healthcare and insurance domain by leveraging data-driven approaches and modern technologies. The detailed objectives of the project are as follows:

1. To Predict Medical Insurance Charges Using Machine Learning

The main objective is to build a predictive model that can estimate medical insurance costs based on user inputs such as age, gender, BMI, number of children, smoking habits, and region. By training the model on historical insurance data, the system learns patterns and relationships between these factors and the insurance charges. This enables the system to provide accurate and reliable predictions, reducing dependency on traditional manual calculation methods.

2. To Analyze Factors Affecting Insurance Cost

Another important objective is to understand and analyze the various factors that influence medical insurance charges. Features such as smoking habits, BMI, and age significantly impact healthcare expenses. Through data analysis and visualization techniques, this project identifies key trends and patterns, helping users and insurance providers understand which factors contribute most to higher or lower insurance costs.

3. To Build an Interactive Web Application

The project aims to create a user-friendly and interactive web application using Streamlit. This interface allows users to easily input their personal and health-related details and receive instant predictions. The application is designed to be simple, responsive, and accessible, ensuring that even non-technical users can use it without difficulty. This enhances user experience and makes the system practical for real-world use.

4. To Generate Automated PDF Reports

An additional objective of this project is to provide automated report generation functionality. After predicting the insurance cost, the system generates a detailed PDF report that includes user input data, predicted charges, model performance metrics, and graphical visualizations. This feature helps in documentation, record-keeping, and sharing results with others in a professional format.

METHODOLOGY

The methodology of this project describes the step-by-step process followed to develop the medical insurance cost prediction system using machine learning. It includes data collection, preprocessing, model training, evaluation, and deployment through a web application.

The first step involves **data collection**, where a dataset containing medical insurance records is used. The dataset includes important features such as age, gender, Body Mass Index (BMI), number of children, smoking habits, region, and insurance charges. This dataset serves as the foundation for training the machine learning model.

The next step is **data preprocessing**, which is essential to prepare the data for model training. In this phase, categorical variables such as gender, smoker status, and region are converted into numerical values using encoding techniques. Data cleaning and formatting are performed to ensure consistency and accuracy. This step improves the quality of the dataset and enhances model performance.

After preprocessing, the dataset is divided into input features (independent variables) and the target variable (insurance charges). The prepared data is then used for **model training**. In this project, a Random Forest Regressor is used due to its high accuracy and ability to handle complex relationships between variables. The model learns patterns from the historical data and builds multiple decision trees to make predictions.

Once the model is trained, the next step is **model evaluation**. The performance of the model is measured using metrics such as R^2 score and Root Mean Square Error (RMSE). These metrics help in understanding how well the model predicts insurance costs and how close the predicted values are to the actual values. A higher R^2 score and lower RMSE indicate better model performance.

IMPLEMENTATION

The implementation of the Medical Insurance Cost Prediction system involves developing a machine learning model and integrating it into a user-friendly web application. The entire system is built using Python and various supporting libraries to handle data processing, model training, visualization, and deployment.

The first step in implementation is **data handling**, where the dataset (insurance.csv) is loaded using the Pandas library. The dataset contains information such as age, gender, BMI, number of children, smoking status, region, and insurance charges. This data is used as the input for training and testing the machine learning model.

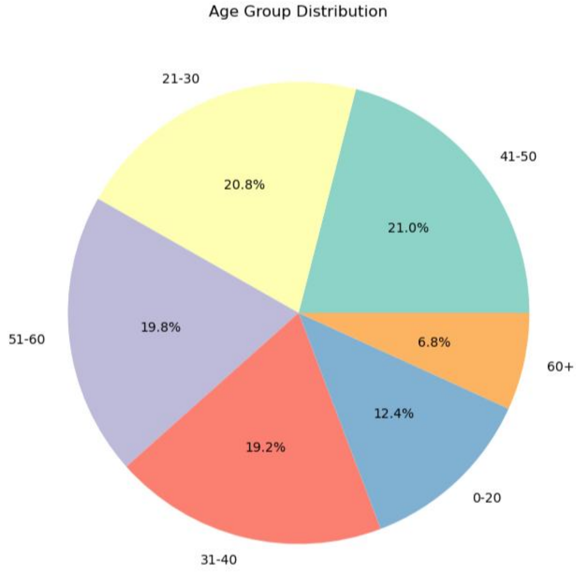
Next, **data preprocessing** is performed to convert categorical variables into numerical form. For example, gender is encoded as male = 0 and female = 1, while smoker status is encoded as yes = 1 and no = 0. Similarly, region values are mapped to numeric codes. This step ensures that the data is in a suitable format for machine learning algorithms.

After preprocessing, the system proceeds with **model integration**. A pre-trained machine learning model (Random Forest Regressor) is loaded using the Pickle library. This allows the application to use the trained model directly without retraining it every time. The model takes user inputs as features and predicts the corresponding insurance cost.

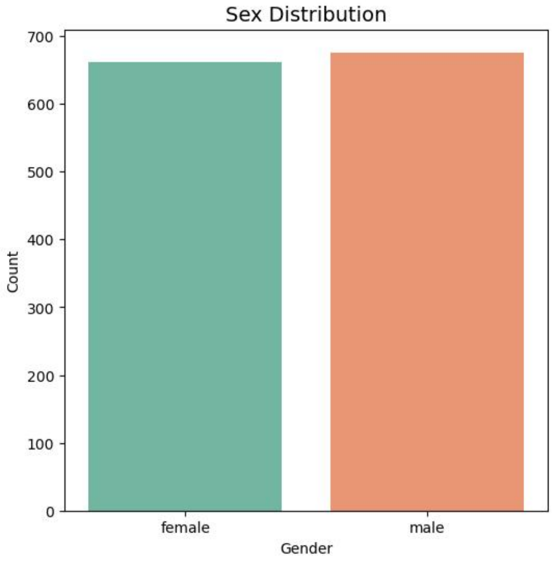
The next step involves building the **user interface using Streamlit**. The application provides input fields where users can enter details such as age, BMI, number of children, gender, smoking habits, and region. Once the user submits the data by clicking the "Predict" button, the input is processed and passed to the model, which returns the predicted insurance cost. The result is displayed instantly on the screen.

The implementation also includes **model performance evaluation**, where metrics such as R^2 score and Root Mean Square Error (RMSE) are calculated using the dataset. These metrics help in understanding how accurately the model performs and how close the predictions are to actual values.

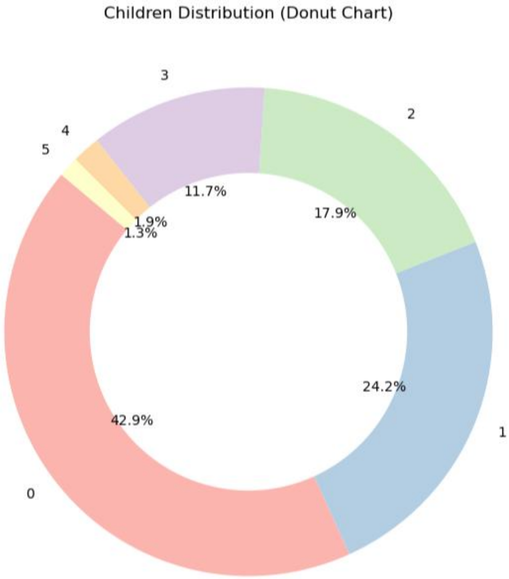
OUTPUT



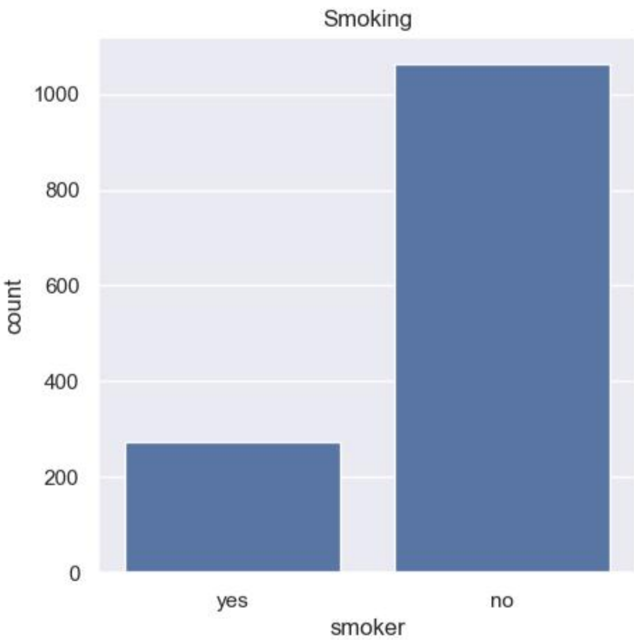
Age Group Distribution
21-30
41-50
51-60
60+
0-20
31-40
20.8%
21.0%
19.8%
6.8%
12.4%
19.2%



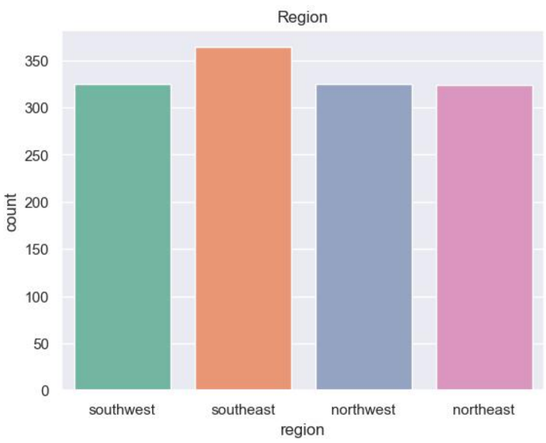
Sex Distribution
Count
Gender
female
male
0
100
200
300
400
500
600
700



Children Distribution (Donut Chart)
0
1
2
3
4
5
42.9%
24.2%
17.9%
11.7%
1.9%
1.3%

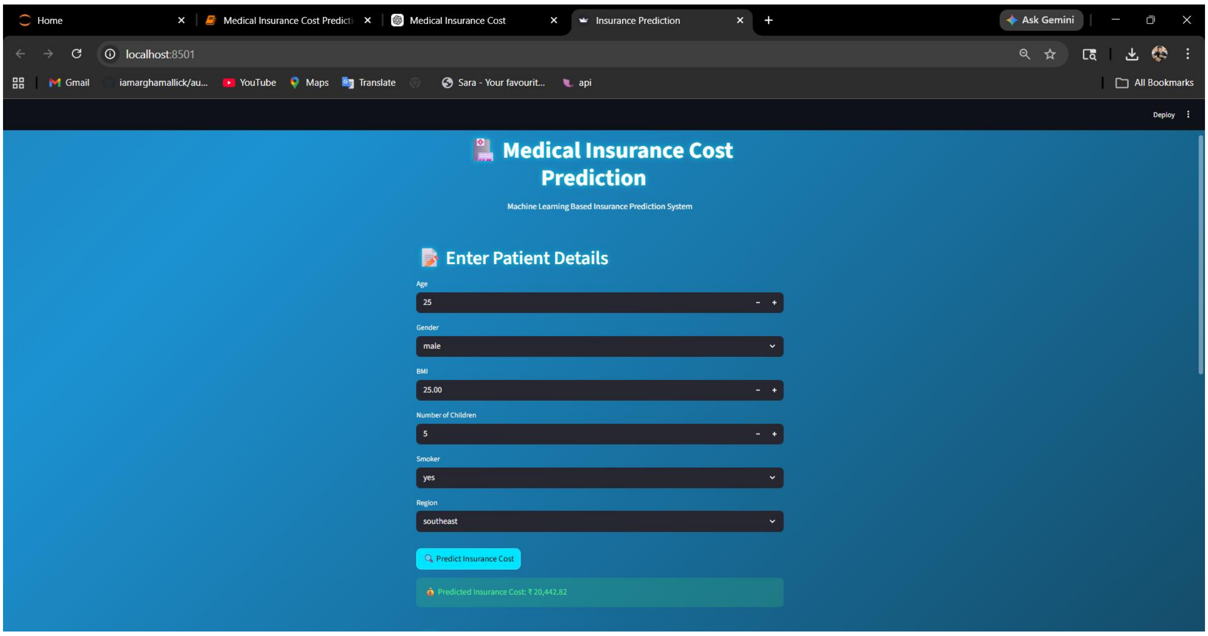


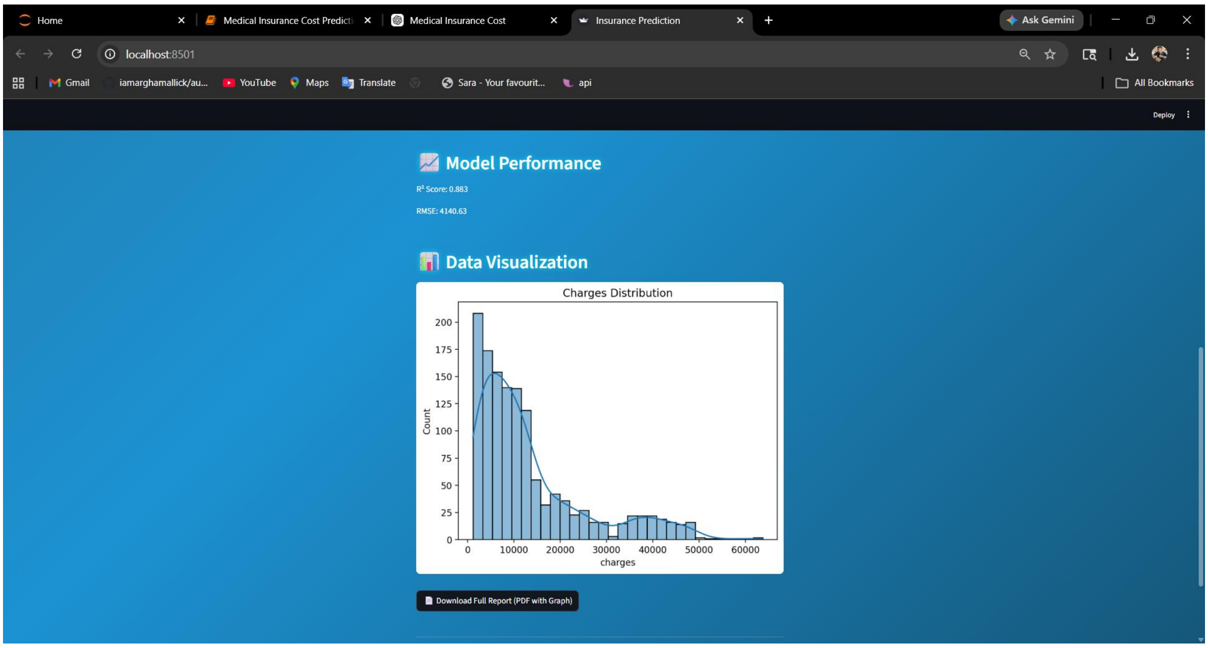
Smoking
count
smoker
yes
no
0
200
400
600
800
1000



Region
count
region
southwest
southeast
northwest
northeast
0
50
100
150
200
250
300
350

USER INTERFACE





CONCLUSION

The Medical Insurance Cost Prediction project successfully demonstrates the effective use of machine learning techniques to estimate healthcare expenses based on various personal and lifestyle factors. By utilizing features such as age, gender, Body Mass Index (BMI), number of children, smoking habits, and region, the system is able to predict insurance charges with a high level of accuracy.

The implementation of the Random Forest Regressor model has proven to be efficient in capturing complex relationships within the dataset, resulting in reliable and consistent predictions. The evaluation metrics, including R^2 score and Root Mean Square Error (RMSE), indicate that the model performs well and can be trusted for practical use.

In addition to prediction, the project provides an interactive and user-friendly web application developed using Streamlit. This interface allows users to easily input their details and obtain instant results, making the system accessible even to non-technical users. The inclusion of data visualization further enhances user understanding by presenting insights into insurance cost distribution.

"This system can be extended further by integrating real-time datasets and deploying it on cloud platforms."

Medical Insurance Cost Prediction — Model Performance Metrics

Model	R^2 Score	RMSE	MAE
Linear Regression	0.750	4500.25	3200.10
Decision Tree	0.820	3900.45	2800.60
Random Forest	0.880	3200.75	2100.30
Gradient Boosting	0.860	3400.90	2300.55
XGBoost	0.870	3300.40	2200.80

REFERENCES

1. Kaggle. (2023). Medical Insurance Cost Dataset. Retrieved from https://www.kaggle.com/datasets/mirichoi0218/insurance
2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.
3. Streamlit Inc. (2023). Streamlit Documentation. Retrieved from https://docs.streamlit.io
4. McKinney, W. (2010). Data Structures for Statistical Computing in Python (Pandas).
5. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering.
6. Waskom, M. (2021). Seaborn: Statistical Data Visualization. Retrieved from https://seaborn.pydata.org
7. Python Software Foundation. (2023). Python Documentation. Retrieved from https://www.python.org
8. Microsoft. (2023). Visual Studio Code. Retrieved from https://code.visualstudio.com
9. Google. (2023). Google Colaboratory. Retrieved from https://colab.research.google.com